Chapter 6. Cycles in Permutations

Not So Vicious Cycles

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A permutation $p = p_1 p_2 \cdots p_n$ of $[n]$ is also the bijection $p : [n] \to [n]$, $p(i) = p_i$. Viewing it as a function, we can compose permutations and follow the orbits of single elements, i.e. the cycles.

Example 6.1

312 is the function f with $f(1) = 3$, $f(2) = 1$, $f(3) = 2$.

The **product** fg of permutations of $[n]$ is the composition "first f, then g": $(fg)(i) = g(f(i))$. (Some authors use the opposite order; this will not matter here.)

Example 6.2

For $f = 312$, $g = 213$: $(fg)(1) = g(3) = 3$, $(fg)(2) = g(1) = 2$, $(fg)(3) = g(2) = 1$, so $fg = 321$.

Example 6.3

For the same f, g: $gf = 132 \neq fg$. Multiplication of permutations is not commutative.

6.1 Cycles in permutations

Motivation. Iterating a permutation moves each element around a closed orbit, so every permutation splits into disjoint cycles. Counting permutations by their cycle structure leads to the Stirling numbers of the first kind, which turn out to be the matrix inverse of those of the second kind.

In $g = 321564$: $g(2) = 2$ is a fixed point, $1 \leftrightarrow 3$ are swapped (a 2-cycle), and $4 \to 5 \to 6 \to 4$ (a 3-cycle).

Lemma 6.4

For a permutation p of $[n]$ and $x \in [n]$, there is $1 \leq i \leq n$ with $p^i(x) = x$.

Proof. Suppose none of $p(x), \ldots, p^n(x)$ equals x. These are n values in the $(n-1)$-element set $[n] \setminus \{x\}$, so by the pigeon-hole principle $p^j(x) = p^k(x)$ for some $1 \leq j < k \leq n$. Applying p^{-1} j times gives $x = p^{k-j}(x)$ with $1 \leq k - j \leq n - 1$, a contradiction. ∎

Definition 6.5 (Cycle)

Let p be a permutation of $[n]$, $x \in [n]$ and i the smallest positive integer with $p^i(x) = x$. Then $x, p(x), \ldots, p^{i-1}(x)$ form an **i-cycle** of p.

Corollary 6.6

Every permutation is the disjoint union of its cycles.

Proof. By Lemma 6.4 every x lies in a cycle. If the cycles of x and y share an element $z = p^a(x) = p^b(y)$, then both cycles are the orbit $\{p^m(z) \mid m \in \mathbb{Z}\}$ of z, so they coincide. Hence distinct cycles are disjoint. ∎

Example 6.7

321564 has the cycles (31), (2), (564).

In cycle notation, $g(i)$ is the entry following i in its cycle (cyclically). The same permutation has many such notations, e.g. $(241)(35) = (53)(412)$. The **canonical cycle form** writes each cycle starting with its largest element and orders the cycles by increasing first elements: $(412)(53)$. The **one-line notation** lists $f(1)f(2)\cdots f(n)$.

Example 6.8

$(412)(53)$ in one-line notation is 24513.

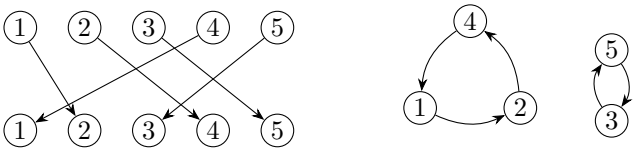


Figure. $24513 = (412)(53)$ *as a function* $[5] \to [5]$ *(left) and as a union of cycles (right).*

Permutations of $[n]$ are called **n-permutations**; their set is the symmetric group S_n.

Theorem 6.9

Let $a_1, \ldots, a_n \geq 0$ with $\sum_i i\, a_i = n$. The number of n-permutations with a_i cycles of length i ($i \in [n]$) is

$$\frac{n!}{a_1!\, a_2! \cdots a_n! \cdot 1^{a_1} 2^{a_2} \cdots n^{a_n}}. \tag{6.1}$$

Proof. Write the elements of $[n]$ in a row ($n!$ ways) and insert parentheses from left to right: a_1 cycles of length 1, then a_2 of length 2, and so on. Every permutation of the given type arises. It arises from exactly $\prod_i i^{a_i} a_i!$ rows: each i-cycle can start at any of its i elements, and the a_i cycles of length i can be listed in any of $a_i!$ orders, independently. ■

$(a_1, \ldots, a_n)$ is the (**cycle**) **type** of the permutation.

Example 6.10

There are $(n-1)!$ n-permutations with a single cycle (type $(0, \ldots, 0, 1)$); equivalently, $(n-1)!$ ways to seat n people around a round table up to rotation.

Definition 6.11 (Stirling numbers of the first kind)

The number of n-permutations with exactly k cycles is the **signless Stirling number of the first kind** $c(n,k)$; $s(n,k) := (-1)^{n-k} c(n,k)$ is the **Stirling number of the first kind**. $c(n,0) = 0$ for $n > 0$, $c(0,0) = 1$, and $c(n,k) = 0$ for $n < k$.

Theorem 6.12

For positive integers $n \geq k$:

$$c(n,k) = c(n-1, k-1) + (n-1)\, c(n-1, k). \tag{6.2}$$

Proof. Either n is a fixed point, and the other $n-1$ entries form $k-1$ cycles: $c(n-1,k-1)$ ways. Or the other entries form k cycles and n is inserted into one of them directly after one of the $n-1$ entries: $(n-1)c(n-1,k)$ ways. Inserting n "in front of" a cycle would give the same cycle as inserting it after the last entry, so these choices are all distinct and exhaustive. ■

Lemma 6.13

For every positive integer n:

$$\sum_{k=0}^{n} c(n,k)x^k = x(x+1)\cdots(x+n-1). \tag{6.3}$$

Proof. Let $G_n(x) := x(x+1)\cdots(x+n-1) = \sum_k a_{n,k}x^k$ (with $G_0 = 1$). From $G_n = (x+n-1)G_{n-1}$,

$$\sum_k a_{n,k}x^k = \sum_k a_{n-1,k-1}x^k + (n-1)\sum_k a_{n-1,k}x^k.$$

Comparing coefficients, $a_{n,k} = a_{n-1,k-1} + (n-1)a_{n-1,k}$: the recurrence (6.2). The initial values also agree ($a_{0,0} = 1$, $a_{n,0} = 0$ for $n > 0$), so $a_{n,k} = c(n,k)$ by induction on n. ■

Replacing x by $-x$ in (6.3) and multiplying by $(-1)^n$:

$$\sum_{k=0}^{n} s(n,k)x^k = (x)_n. \tag{6.4}$$

This explains the sign in $s(n,k)$. Compare with (5.2): $x^n = \sum_k S(n,k)(x)_k$. In the vector space of real polynomials, both $B = \{1, x, x^2, \dots\}$ and $B' = \{1, (x)_1, (x)_2, \dots\}$ are bases ($(x)_k$ has degree k). Let S and s be the infinite lower-triangular matrices with entries $S(n,k)$, $s(n,k)$; then (5.2) and (6.4) say that S and s are the two change-of-basis matrices.

Theorem 6.14

The matrices S and s are inverse to each other: $Ss = sS = I$.

Proof. Both matrices are lower triangular, so all products below are finite sums. Substituting (6.4) into (5.2),

$$x^n = \sum_k S(n,k)(x)_k = \sum_j x^j \sum_k S(n,k)s(k,j).$$

Since the monomials are linearly independent, $\sum_k S(n,k)s(k,j) = \delta_{nj}$, i.e. $Ss = I$. Substituting (5.2) into (6.4) gives $(x)_n = \sum_j (x)_j \sum_k s(n,k)S(k,j)$. The $(x)_j$ are linearly independent, so $sS = I$. ■

6.2 Permutations with restricted cycle structure

Motivation. Erasing the parentheses of the canonical cycle form gives a bijection $S_n \to S_n$ that turns statements about cycles into statements about positions. With it, probabilities about cycles become transparent, and it leads to the surprising square formula for permutations with only odd (or only even) cycles.

Lemma 6.15 (Transition lemma)

For $p \in S_n$ in canonical cycle form, let $g(p)$ be the word obtained by erasing the parentheses, read in one-line notation. Then $g : S_n \to S_n$ is a bijection.

Example 6.16

$g((412)(53)) = 41253$.

Proof of Lemma 6.15. We show that every $q = q_1 \cdots q_n$ admits exactly one way of inserting parentheses that yields a canonical cycle form. In a canonical form, each cycle starts with its maximum and the leading entries increase. So a new cycle starts at q_k iff q_k is larger than all leading entries before it. Since each leading entry is the maximum of its cycle, this happens iff q_k is larger than *all* entries to its left, i.e. q_k is a **left-to-right maximum**. Hence the cycle starts, and so the parentheses, are forced, and they produce a valid canonical form. ■

Example 6.17

$g^{-1}(4356172) = (43)(5)(61)(72)$: the left-to-right maxima $4, 5, 6, 7$ open the cycles.

If q has t left-to-right maxima then $g^{-1}(q)$ has t cycles; q_1 and the entry n are always left-to-right maxima.

Proposition 6.18

For $i \neq j$ in $[n]$, exactly half of all n-permutations contain i and j in the same cycle.

Proof. Relabelling ($i \leftrightarrow n$, $j \leftrightarrow n-1$) preserves cycle structure, so let $i = n$, $j = n-1$. In $q = g(p)$ the entry n is the last left-to-right maximum, so it starts the last cycle of p. That cycle consists of n and the entries to its right in q. Hence $n, n-1$ share a cycle iff $n-1$ is to the right of n in q, which holds for half of all q (swap the two entries). Since g is a bijection, the claim follows. ■

Lemma 6.19

For $i \in [n]$ and every $k \in [n]$, exactly $(n-1)!$ n-permutations have i in a cycle of length k.

Proof. By relabelling let $i = n$. If $q = g(p)$ has $q_j = n$, the cycle of n in p is $q_j \cdots q_n$, of length $n - j + 1$. So its length is k iff n is in position $n + 1 - k$ of q, which holds for $(n-1)!$ words q. ■

So the probability that a given entry lies in a k-cycle is $\frac{1}{n}$, independently of k.

Let ODD(m), resp. EVEN(m), be the m-permutations all of whose cycles have odd, resp. even, length.

Lemma 6.20

$|\mathrm{ODD}(2m)| = |\mathrm{EVEN}(2m)|$ for all positive integers m.

Proof.

Step 1 (the map Φ). Let $\pi \in \mathrm{ODD}(2m)$. Since $2m$ is even, π has an even number $2k$ of odd cycles $C_1, \ldots, C_{2k}$ (canonical order). For each $i \leq k$ move the last element of C_{2i-1} to the end of C_{2i}; a singleton C_{2i-1} disappears. The lengths become $|C_{2i-1}| - 1$ and $|C_{2i}| + 1$, both even. The moved element is smaller than the leading entry of C_{2i-1} (or is that entry, for a singleton), hence smaller than the leading entry of C_{2i}. So canonical form is preserved and $\Phi(\pi) \in \mathrm{EVEN}(2m)$.

Step 2 (the inverse). Let $\sigma \in \mathrm{EVEN}(2m)$ have canonical cycles $c_1, \ldots, c_h$. Process from the right. If the last entry of c_h exceeds the first entry of c_{h-1}, it cannot have come from c_{h-1}, so it was a singleton cycle placed just before c_h: restore it, and continue with c_{h-1} as the new last cycle. Otherwise move it back to the end of c_{h-1} and continue with c_{h-2}. If a single cycle remains, its last entry was a singleton. Each step is forced by Step 1, so the preimage exists and is unique. ■

Example 6.21

$\Phi((4)(513)(726)(8)) = (5134)(72)(86)$.

Example 6.22

$\Phi^{-1}((41)(62)(75)(83)) = (412)(6)(753)(8)$.

Example 6.23

$\Phi^{-1}((21)(53)(64)(87)) = (1)(2)(534)(6)(7)(8)$.

Theorem 6.24

For every positive integer m:

$$|\mathrm{ODD}(2m)| = |\mathrm{EVEN}(2m)| = 1^2 \cdot 3^2 \cdot 5^2 \cdots (2m-1)^2. \tag{6.5}$$

Proof. By Lemma 6.20 it suffices to count $\mathrm{EVEN}(2m)$. Build p by following cycles: start with 1, choose $p(1)$, $p^2(1)$, ..., and whenever a cycle closes, continue with the smallest unused element.

Step 1. $p(1) \neq 1$ (no 1-cycles): $2m-1$ choices. $p^2(1)$: anything except $p(1)$, so again $2m-1$ choices; choosing 1 closes a 2-cycle.

Step 2. In general, when choosing the $(2i-1)$-st image, $2i-2$ elements are used and the current cycle has even length or is just starting. The image may be neither used nor the current start element (an odd cycle, possibly of length 1, would close): $2m-2i+1$ choices. When choosing the $2i$-th image, the current cycle has odd length, so closing it to even length is allowed: the image is any of the $2m-(2i-1)$ elements that are not yet images, again $2m-2i+1$ choices.

Step 3. Multiplying, $|\mathrm{EVEN}(2m)| = \prod_{i=1}^{m}(2m-2i+1)^2$. ■

Of course $\mathrm{EVEN}(n) = \emptyset$ for odd n, but $\mathrm{ODD}(n)$ can still be counted.

Theorem 6.25

For every positive integer m:

$$|\mathrm{ODD}(2m+1)| = (2m+1)\,|\mathrm{ODD}(2m)| = 1^2 \cdot 3^2 \cdots (2m-1)^2\,(2m+1). \tag{6.6}$$

Proof. By Lemma 6.27 below, Ψ is a bijection $\mathrm{ODD}(2m) \times [2m+1] \to \mathrm{ODD}(2m+1)$; combine with Theorem 6.24. ■

An m-permutation in cycle form has $m+1$ **gap positions**: one at the very beginning and one after each entry.

Example 6.26

$(42)(513)$ has six gap positions: $|(4|2|)(5|1|3|)$.

The map Ψ. For $\pi \in \mathrm{ODD}(2m)$ and $k \in [2m+1]$: take $\Phi(\pi) \in \mathrm{EVEN}(2m)$, add 1 to each entry (giving a permutation of $\{2, \dots, 2m+1\}$), and insert the new entry 1 into the k-th gap position. This makes one cycle odd and keeps the canonical form (1 is never a leading entry, and inserting it at the very front puts it in a new fixed point (1) first). Finally apply Φ^{-1} (with relabelling) to the remaining, all-even cycles. The result $\Psi(\pi, k)$ has only odd cycles.

Lemma 6.27

Ψ is a bijection from $\mathrm{ODD}(2m) \times [2m+1]$ onto $\mathrm{ODD}(2m+1)$.

Proof. Given $\pi' \in \mathrm{ODD}(2m+1)$, set aside the cycle containing 1 and apply Φ to the remaining cycles (an even number of elements, all odd cycles) to get even cycles. Put the cycle of 1 back and read off k as the gap position of 1. Delete 1; the cycle of 1 becomes even, so the result lies in EVEN, on $\{2, \dots, 2m+1\}$. Subtract 1 from all entries and apply Φ^{-1} to get π. Every step inverts the corresponding step of Ψ, so Ψ is invertible. ■

6.3 Selected exercises

Motivation. The book exercises below use the cycle decomposition in three ways: to set up a recurrence, to average a statistic with Lemma 6.19, and to decide which permutations are squares.

Exercise 6.5 (Involutions)

Let $r(n)$ be the number of n-permutations p with $p^2 = \mathrm{id}$, and $r(0) = 1$. Prove that $r(n+1) = r(n) + n\, r(n-1)$ for $n \geq 1$.

Solution. $p^2 = \mathrm{id}$ means that every cycle of p has length 1 or 2. Split the involutions p of $[n+1]$ by the cycle containing $n+1$.

Step 1 ($n+1$ is fixed). Then p restricted to $[n]$ is an involution of $[n]$, and every involution of $[n]$ extends in this way by $p(n+1) = n+1$. This gives $r(n)$ involutions.

Step 2 ($n+1$ is in a 2-cycle). Then p contains $(k\ n{+}1)$ for exactly one $k \in [n]$, and the other cycles form an involution of the $(n-1)$-element set $[n] \setminus \{k\}$. Conversely each k and each such involution give one p. This gives $n\, r(n-1)$ involutions (for $n = 1$, $r(0) = 1$ accounts for the empty involution).

Adding the two cases, $r(n+1) = r(n) + n\, r(n-1)$. ■

Exercise 6.7 (The average number of cycles)

Prove that n-permutations have on average $H_n = \sum_{i=1}^{n} \frac{1}{i}$ cycles.

Solution.

Step 1 (cycles as a sum over entries). For $p \in S_n$ and $i \in [n]$ let $\ell_p(i)$ be the length of the cycle of p containing i. A cycle of length ℓ contains ℓ entries, each contributing $\frac{1}{\ell}$, so the number of cycles of p is

$$\mathrm{cyc}(p) = \sum_{i=1}^{n} \frac{1}{\ell_p(i)}.$$

Step 2 (double counting). Summing over all p and exchanging the sums,

$$\sum_{p \in S_n} \mathrm{cyc}(p) = \sum_{i=1}^{n} \sum_{k=1}^{n} \frac{1}{k} \left|\{p : \ell_p(i) = k\}\right| = \sum_{i=1}^{n} \sum_{k=1}^{n} \frac{(n-1)!}{k} = n!\, H_n,$$

where Lemma 6.19 gives $|\{p : \ell_p(i) = k\}| = (n-1)!$.

Step 3 (conclusion). Dividing by $|S_n| = n!$, the average number of cycles is H_n. ■

Exercise 6.21 (Square roots of permutations)

A permutation $p \in S_n$ has a *square root* if $q^2 = p$ for some $q \in S_n$. Prove that p has a square root iff, for every even m, the number of cycles of p of length m is even.

Solution.

Step 1 (square of one cycle). Let $c = (x_0\, x_1 \cdots x_{\ell-1})$, i.e. $c(x_i) = x_{i+1}$ with indices modulo ℓ. Then $c^2(x_i) = x_{i+2}$. If ℓ is odd, the indices $0, 2, 4, \ldots$ run through all residues modulo ℓ before returning to 0 (since 2 is invertible modulo ℓ), so $c^2 = (x_0\, x_2\, x_4 \cdots)$ is a single ℓ-cycle. If ℓ is even, $c^2 = (x_0\, x_2 \cdots x_{\ell-2})(x_1\, x_3 \cdots x_{\ell-1})$ consists of two cycles of length $\ell/2$.

Step 2 (necessity). Let $q^2 = p$. Each cycle C of q is mapped to itself by q, and q^2 restricted to C is the square of that cycle. So the cycles of p arise from the cycles of q as in Step 1. An odd cycle of q yields one odd cycle, so a cycle of p of even length m comes from a cycle of q of length $2m$, and that cycle yields exactly two cycles of length m. Hence the m-cycles of p come in pairs, and their number is even.

Step 3 (sufficiency). Assume the condition. Pair up the cycles of each even length arbitrarily, and treat each odd cycle separately. For an odd cycle c of length $\ell = 2r+1$, let $q = c^{r+1}$ on its elements; then $q^2 = c^{2r+2} = c^\ell c = c$ there. For a pair $(y_0 \cdots y_{m-1})$, $(z_0 \cdots z_{m-1})$ of m-cycles, let q be the $2m$-cycle $(y_0\, z_0\, y_1\, z_1 \cdots y_{m-1}\, z_{m-1})$ on their elements; then $q^2(y_i) = y_{i+1}$ and $q^2(z_i) = z_{i+1}$, so q^2 is the given pair there.

Step 4 (conclusion). The pieces of Step 3 live on disjoint sets covering $[n]$, so together they define $q \in S_n$, and $q^2 = p$ on every piece, hence everywhere. ■